PROGRAMMING II: OBJECT-ORIENTED PROGRAMMING (2025)

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PROJECT INSTRUCTIONS

The programming II – course (7 CR) includes a project. The weekly exercises and the exam cover 5ECTS, while the project covers 2ECTS. The project consists of a small but functional application, written in Java. A project document must also be written.

MINIMUM REQUIREMENTS

OBJECT-ORIENTATION (OOP): The project must make proper use of encapsulation (classes and objects), with properly designed classes (with proper visibility modifiers, constructors, and other class members), including inheritance (subclass and superclass relationship). The project may also include other OOP features (e.g. abstract classes, interfaces and/or polymorphism). The project must include a minimum of 5 Java classes, abstract classes, or interfaces, excluding JavaFX classes. (for example: the classes Artwork, Song, Book, Painting count to 4 classes).

DATA AND FILES: The project must handle data in one way or the other. Usually this means reading or writing text files, or object serialization. This means that the application should be able to save and/or load data.

A GRAPHICAL USER INTERFACE IN JAVAFX: The project must include a proper GUI, which must be self-coded in JavaFX (and not generated by a user interface designer program). The GUI does not need to be complex; simple functionality is enough. Proper UI elements and related event handlers and/or listeners must be in place.

DOCUMENTATION: The project must be properly documented.

PEER-REVIEW: Review two projects by fellow students.

TIMETABLES AND DEADLINES

- Choose project topic by: 1st of March
- Project deadline: 31st of March (23:59)
- Peer-review deadline: 7th of April (23:59)

SELECTING THE TOPIC

You can freely select a topic of your own preference. But two projects can't be on the same topic, unless they differ significantly in their approach. The topic suggestion must be posted to the e-Learn discussion forum of the course (a specific area will be opened for this purpose: check that your topic is not already taken). Keep the project as simple as possible. A simple but well-functional project is better than a large but dysfunctional one. When thinking about

the topic, think about what Java classes the project might have, how the UI might look like, and how the project could use files and/or serialization. Some example ideas:

- **A simple library system for books.** Should include at minimum the possibility to list books, add books, perhaps some basic search. Classes could include Books, Authors, Genres. What about inheritance: subclasses and superclasses? Maybe one class for managing the books (perhaps in an ArrayList structure?) including the file operations and/or serialization?
- **A recipe manager**
- **A game, such as a maze game, card game, or arcade game**
- **Wardrobe manager**
- **Artwork collection**
- **Restaurant ordering system**
- **A system for taking notes**
- **A system for arranging multiple choice questionnaires**
- **... or anything else**

SUBMITTING THE PROJECT

The project is submitted as one zip-file, which contains all project codes and other necessary files to run the project, and the project document as one **PDF** file.

PROJECT DOCUMENT

The project document should include:

- Title and author
- General description. What is the purpose of the project, and who are the potential users of the software? (Minimum: 2 paragraphs)
- Screenshots of the UI.
- A visualization and/or a clear textual description of the Java – classes used in the project. Explain the purpose of each class.
- Brief story of project evolution (Minimum: 2 paragraphs)
- Usage instructions for a novice user. (Minimum: 1 paragraph)
- Self-evaluation of the project. What are the positive aspects of the project? What about challenges or problems? (Minimum: 1 paragraph)
- Further development. How could / should the project be developed? (Minimum: 1 paragraph)

EVALUATION CRITERIA

Success criteria:

- Proper use of OOP (+)
- Proper implementation of a UI (+)
- The program works properly without major problems (+)
- Proper commenting of code (+)
- The documentation covers all required parts (+)
- The project is written solely by the author (if some part of the code is not written by the author, this is clearly indicated) (+)

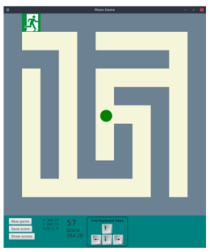
Failure criteria:

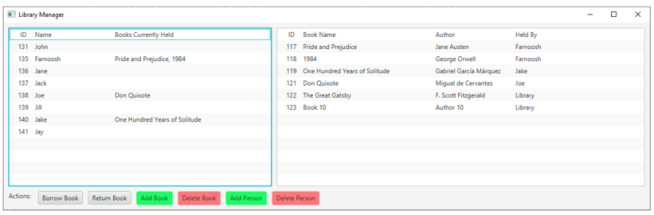
- OOP principles are not used (-)
- The project or a part of the project has been copied and/or implemented by someone else than the author (and this is not clearly indicated in the documentation) (-)
- Program does not work at all or constantly fails or crashes (-)
- The project does not include a functional UI (-)
- The project can't handle files or serialization (-)
- The project is not documented (-)

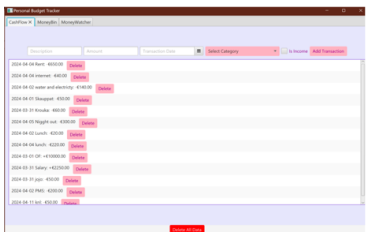
Evaluation scale: fail/pass (an excellent project that fulfills all criteria earns +1 to final grade)

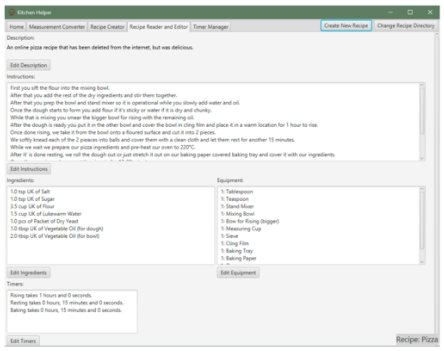
EXAMPLES FROM LAST YEAR

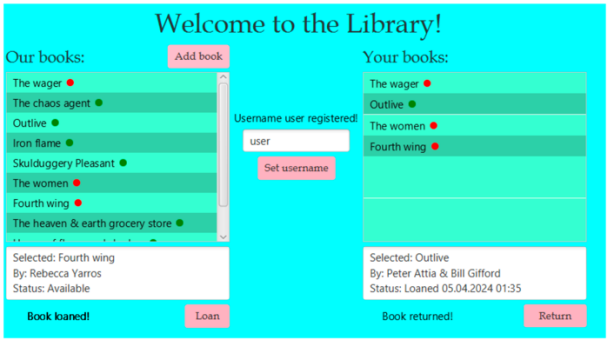
Here are some screenshots from last year projects for inspiration. Some of these projects are very advanced and beyond the requirements. Remember: a simple but functional project is enough. Also: a simple but functional program is better than a complex but failing one.

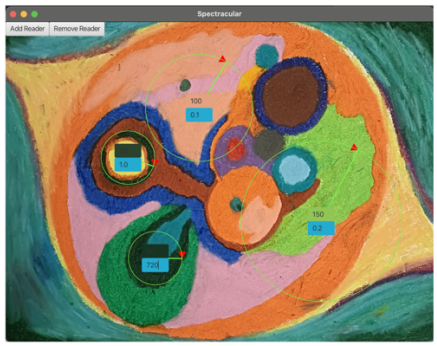


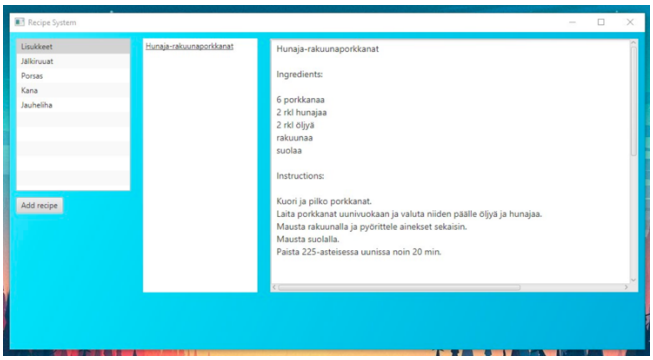


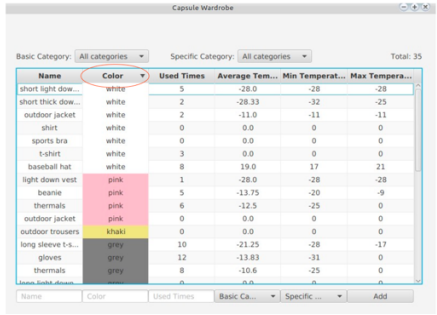


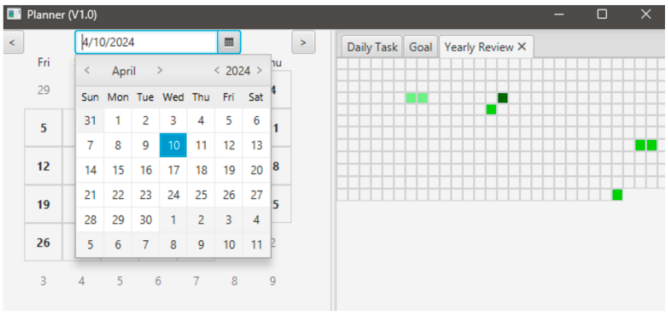

Kitchen Helper
Home
Measurement Converter
Recipe Creator
Recipe Reader and Editor
Timer Manager
Create New Recipe
Change Recipe Directory
Description:
An online pizza recipe that has been deleted from the internet, but was delicious.
Edit Description
Instructions:
First you sift the flour into the mixing bowl.
After that you add the rest of the dry ingredients and stir them together.
After that you prep the bowl and stand mixer so it is operational while you slowly add water and oil.
Once the dough starts to form you add flour if it's sticky or water if it is dry and chunky.
While that is mixing you smear the bigger bowl for rising with the remaining oil.
After the dough is ready you put it in the other bowl and cover the bowl in cling film and place it in a warm location for 1 hour to rise.
Once done rising, we take it from the bowl onto a floured surface and cut it into 2 pieces.
We softly knead each of the 2 peaces into balls and cover them with a clean cloth and let them rest for another 15 minutes.
While we wait we prepare our pizza ingredients and pre-heat our oven to 220°C.
After it' is done resting, we roll the dough out or just stretch it out on our baking paper covered baking tray and cover it with our ingredients.
Edit Instructions
Ingredients:
1.0 tsp UK of Salt
1.0 tsp UK of Sugar
3.5 cup UK of Flour
1.5 cup UK of Lukewarm Water
1.0 pcs of Packet of Dry Yeast
1.0 tbsp UK of Vegetable Oil (for dough)
2.0 tbsp UK of Vegetable Oil (for bowl)
Edit Ingredients
Equipment:
1: Tablespoon
1: Teaspoon
1: Stand Mixer
1: Mixing Bowl
1: Bow for Rising (bigger)
1: Measuring Cup
1: Sieve
1: Cling Film
1: Baking Tray
1: Baking Paper
Edit Equipment
Timers:
Rising takes 1 hours and 0 seconds.
Resting takes 0 hours, 15 minutes and 0 seconds.
Baking takes 0 hours, 15 minutes and 0 seconds.
Edit Timers
Recipe: Pizza


Welcome to the Library!
Our books:
Add book
The wager
The chaos agent
Outlive
Iron flame
Skulduggery Pleasant
The women
Fourth wing
The heaven & earth grocery store
Username user registered!
user
Set username
Your books:
The wager
Outlive
The women
Fourth wing
Selected: Fourth wing
By: Rebecca Yarros
Status: Available
Book loaned!
Loan
Selected: Outlive
By: Peter Attia & Bill Gifford
Status: Loaned 05.04.2024 01:35
Book returned!
Return

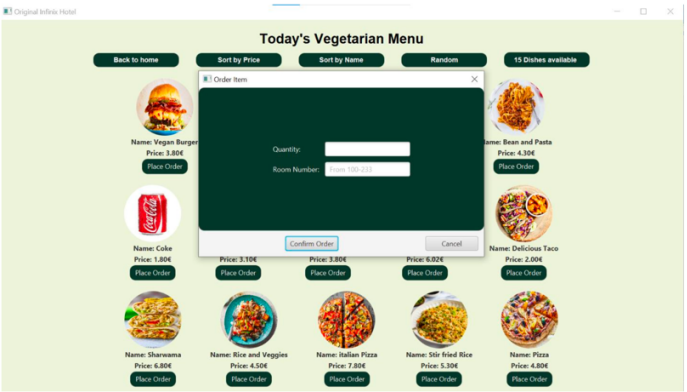

Spectracular
Add Reader
Remove Reader
100
0.1
1.0
150
0.2
720

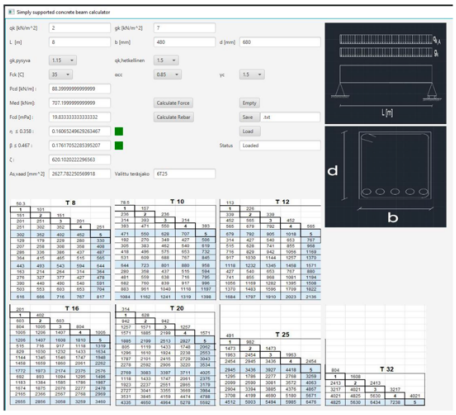

Recipe System
Lisukkeet
Jälkiruuat
Porsas
Kana
Jauheliha
Add recipe
Hunaja-rakuunaporkkanat
Hunaja-rakuunaporkkanat
Ingredients:
6 porkkanaa
2 rkl hunajaa
2 rkl öljyä
rakuunaa
suolaa
Instructions:
Kuori ja pilko porkkanat.
Laita porkkanat uunivuokaan ja valuta niiden päälle öljyä ja hunajaa.
Mausta rakuunalla ja pyörittele ainekset sekaisin.
Mausta suolalla.
Paista 225-asteisessa uunissa noin 20 min.

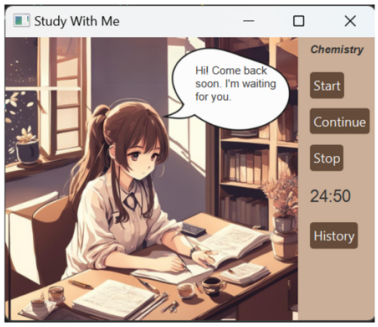

Capsule Wardrobe
Basic Category: All categories
Specific Category: All categories
Total: 35
Name
Color
Used Times
Average Tem...
Min Temperat...
Max Tempera...
short light dow... white 5 -28.0 -28 -28
short thick dow... white 2 -28.33 -32 -25
outdoor jacket white 2 -11.0 -11 -11
shirt white 0 0.0 0 0
sports bra white 0 0.0 0 0
t-shirt white 3 0.0 0 0
baseball hat white 8 19.0 17 21
light down vest pink 1 -28.0 -28 -28
beanie pink 5 -13.75 -20 -9
thermals pink 6 -12.5 -25 0
outdoor jacket pink 0 0.0 0 0
outdoor trousers khaki 0 0.0 0 0
long sleeve t-s... grey 10 -21.25 -28 -17
gloves grey 12 -13.83 -31 0
thermals grey 8 -10.6 -25 0
Name
Color
Used Times
Basic Ca...
Specific ...
Add

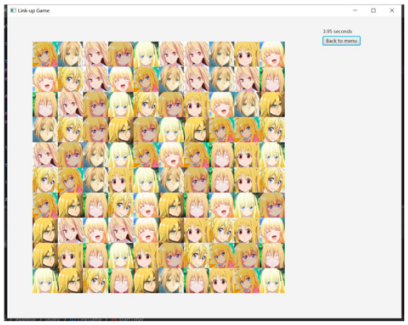

Planner (V1.0)
4/10/2024
April
2024
Sun Mon Tue Wed Thu Fri Sat
Daily Task
Goal
Yearly Review

MEMO. Time: 11 seconds
Sam: 0
CURRENT PLAYER: Den
Den: 0


Original Infinix Hotel
Today's Vegetarian Menu
Back to home
Sort by Price
Sort by Name
Random
15 Dishes available
Order Item
Quantity:
Room Number:
Confirm Order
Cancel
Name: Vegan Burger
Price: 3.80€
Place Order
Name: Bean and Pasta
Price: 4.30€
Name: Coke
Price: 1.80€
Name: Delicious Taco
Price: 2.00€
Name: Sharwama
Price: 6.80€
Name: Rice and Veggies
Price: 4.50€
Name: Italian Pizza
Price: 7.80€
Name: Stir fried Rice
Price: 5.30€
Name: Pizza
Price: 4.80€


Simply supported concrete beam calculator
T 8
T 10
T 12
T 16
T 20
T 25
T 32


Study With Me
Hi! Come back soon. I'm waiting for you.
Chemistry
Start
Continue
Stop
24:50
History


Link-up Game
3.95 seconds
Back to menu